

Windows Registry Monitoring System

Designed & Implemented by Kumar Ashwar

A Forensics-Grade Persistence Detection & Integrity Toolkit

Project Overview

An advanced Windows utility focused on tracking unauthorized changes, identifying malicious persistence, and verifying system integrity using Python-based baseline comparisons.

| Core Project Objectives



Persistence Detection

Monitor sensitive autorun keys to identify malware trying to survive system reboots.



Integrity Auditing

Implement a baseline comparison engine to detect unauthorized additions or deletions.



Real-time Alerts

Generate timestamped logs and instant alerts for suspicious registry modifications.



Practical Motivation

Malware often targets the registry as its "hidden" home. The registry is a high-value target for:

- ! Ransomware configuration changes
- ! Unauthorized startup entries
- ! Privilege escalation modifications
- ! Disabling security tools like Defender

| Practical Scope

Autorun Key Monitoring

Focus on **Run** and **RunOnce** keys in HKCU and HKLM hives. Tracks executables launched during user login and system boot.

Malware-Like Changes

Monitoring security policies, firewall/defender status, and UAC bypass keys to detect active tampering attempts.

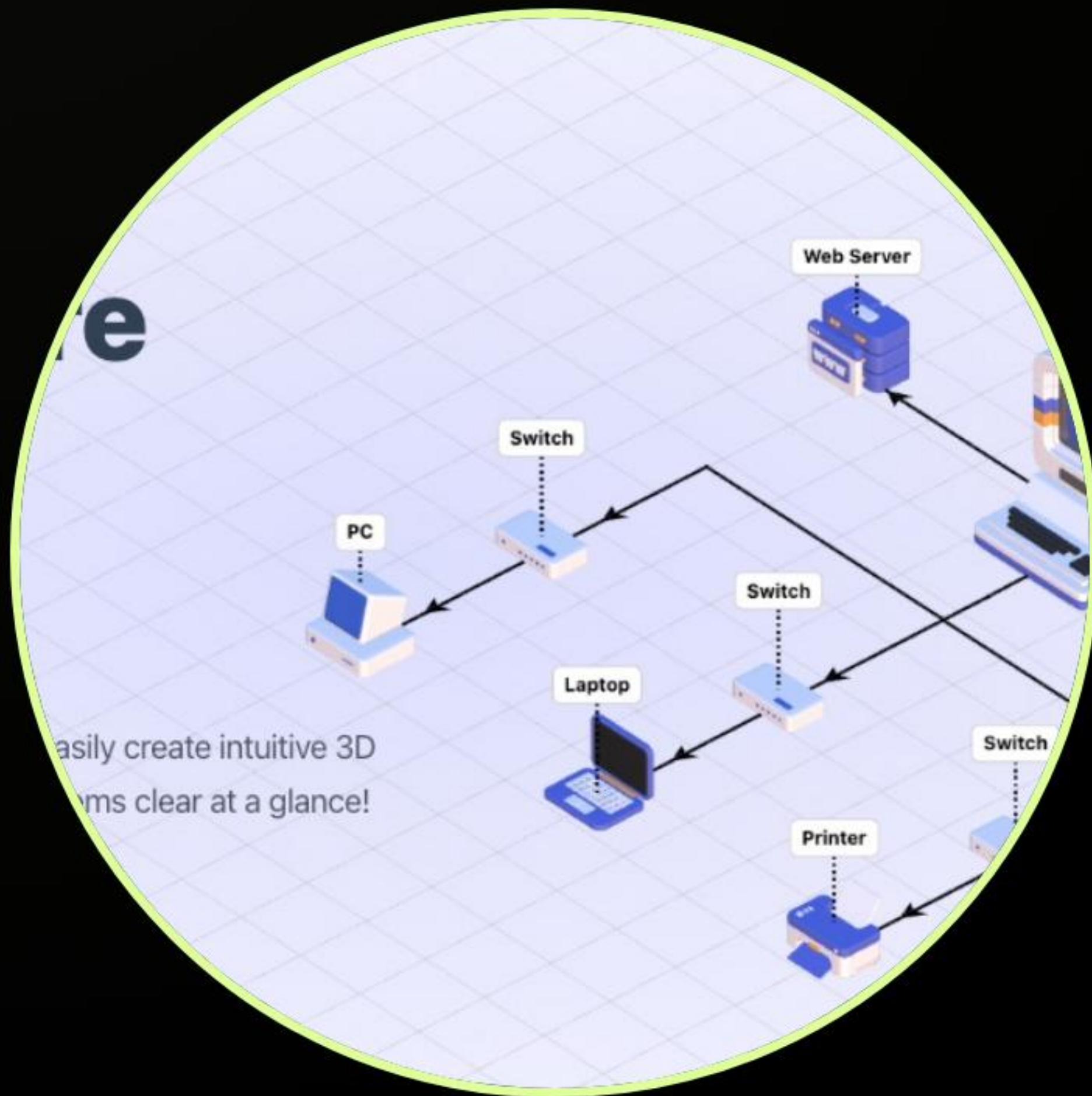
| Tools & Technologies

-  **Python & Winreg:** Core language and library for deep registry access and enumeration.
-  **PowerShell:** Used for optional manual verification and advanced item properties.
-  **JSON & Hashlib:** Baseline storage using JSON and integrity verification via checksums.
-  **Task Scheduler:** Automating periodic scans for continuous monitoring.

| Workflow Architecture



Architecture Overview



System Logic

The system utilizes `monitor.py` to iterate through defined hives. It cross-references current keys against the `registry_baseline.json` state. Discrepancies are flagged as New Additions, Deletions, or Modifications, providing a granular look at registry health.

Key Monitoring Checklist

Registry Path (Hive)	Monitoring Purpose	Risk Level
HKLM\...\Run	System-wide startup persistence	Critical
HKCU\...\Run	User-specific startup apps	High
HKLM\Policies\Microsoft	Security tools & policies	Critical
HKLM\...\CurrentVersion\RunOnce	Temporary malware installers	Medium

| Prevalence of Persistence Techniques



Market Data: Registry-based persistence remains the dominant method for malware installers due to ease of implementation and high reliability.



Expected Output

The toolkit outputs real-time timestamped logs. Example: [2026-07-04] New Entry Detected: C:\Users\User\AppData\malware.exe. It produces a detailed forensics report and alerts for any integrity failures from the baseline state.

Thank You

Questions & Feedback

Kumar Ashwar | Windows Registry Monitoring Project

| Image Sources



https://cdn.prod.website-files.com/63f5c98060519d42bb6be4a1/698055410946938a8936b2bf_what-is-cybersecurity-awareness-training-cybersecurity.jpeg

Source: www.getcourse.co.nz



<https://raw.githubusercontent.com/gantFDT/icraft/main/public/images/banner.jpg>

Source: github.com



https://elements-resized.envatousercontent.com/elements-cover-images/abf26ea1-bf16-41a9-9ee7-57b3f52132d5?w=433&cf_fit=scale-down&q=85&format=auto&s=5627d71ee99148745fb7adc0ac9d2c4538fc103edele75124453ac78314de9aa

Source: elements.envato.com
